# A local noise tradeoff for two cross-regulated genes with unequal finite RNA lifetimes

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**Abstract**

Eliminating short-lived RNA gives an instantaneous protein-burst model, but a noise constraint in that limit need not describe finite RNA lifetimes. We establish a computer-assisted local constraint for two genes with explicit RNA, catalytic translation and protein degradation. In protein-lifetime units, the constitutive transcription and RNA-decay rates are respectively $(1, 1)$ and $(2, 2)$, while both translation and protein-decay coefficients are one. Each parent has RNA and protein mean one. Cross-transcription responses may be any bounded real functions of the other protein count, including nonmonotone functions and responses confined to arbitrarily rare counts. An infinite-state return-operator bound, obtained from weighted generator estimates and exact finite Riesz residual certificates, is combined with a controlled cubic covariance remainder. For each fixed nonconstant response pair, an explicit positive feedback interval forces at least one stationary protein Fano factor above its own mean-matched finite-lifetime baseline, $3/2$ or $4/3$. Constant responses leave both baselines unchanged. The interval depends on the response shapes. The result does not settle strong feedback, other kinetic parameters or networks with more genes.

## 1 Finite RNA memory and output-noise constraints

At low molecule counts, a gene-regulatory circuit is naturally represented by stochastic reaction events. Noise control must then be compared with an uncontrolled system at the same stationary mean, rather than with an arbitrarily chosen production rate. For explicit RNA and catalytic translation, the constitutive protein Fano factor is

$$B_i = 1 + \frac{\beta_i}{\delta_i + \gamma_i}, \tag{1}$$

where $\delta_i$ and $\gamma_i$ are RNA and protein decay coefficients and $\beta_i$ is the translation coefficient [1, 2]. This finite-lifetime baseline differs from the instantaneous-burst value $1 + \beta_i/\delta_i$.

Yan *et al.* formulated noise-tradeoff conjectures for reaction networks and established analytical constraints in linear-noise and separated-time regimes [3]. Stationary information-flow arguments subsequently supplied exact constraints for unit-birth components relative to their Poisson baselines [4, 5]. An exact tradeoff also holds for instantaneous positive geometric protein bursts, while unrestricted nongeometric burst laws admit counterexamples [6]. The latter work identifies finite-lived production intermediates as a remaining direction. A protein-only burst generator omits the RNA memory retained here. Likewise, a constraint on all RNA and protein coordinates relative to one does not by itself compare only the proteins with Eq. (1).

We consider a restricted, exactly specified part of that direction: two genes with unequal finite RNA lifetimes and low constitutive mean counts. The contribution is a local output-noise obstruction

valid for every bounded response shape on the complete count lattice. The response functions are not linearized at their means. The proof combines exact stationary identities, a computer-assisted bound on the complete response-function space, and an explicit finite-amplitude remainder. The fixed kinetics and response-dependent amplitude are essential parts of the statement.

## 2 Reaction model and the exact comparison

Let $(M_1, P_1, M_2, P_2) \in \mathbb{Z}^4_{\geq 0}$ denote RNA and protein counts. For $i = 1, 2$, with $j \neq i$, the only channels are

$$\begin{aligned} M_i &\longrightarrow M_i + 1 \quad \text{at rate } f_i(P_j), & M_i &\longrightarrow M_i - 1 \quad \text{at rate } \delta_i M_i, \\ P_i &\longrightarrow P_i + 1 \quad \text{at rate } \beta_i M_i, & P_i &\longrightarrow P_i - 1 \quad \text{at rate } \gamma_i P_i. \end{aligned} \tag{2}$$

Translation leaves RNA unchanged. Death rates vanish at zero, and there is no upper count boundary. Transcription depends only on the other protein: there is no direct protein self-regulation, promoter state, degradation feedback, extrinsic environment or additional controller. No detailed-balance assumption is imposed.

Choose a common time unit $t_0$ equal to the protein mean lifetime. All times and rates below are their numerical values in units $t_0$ and $t_0^{-1}$. Counts, response functions and $\epsilon$ are dimensionless; $a, b$ are numerical transcription gains in the same rate unit. The fixed parameters are

$$(\delta_1, \beta_1, \gamma_1) = (1, 1, 1), \qquad (\delta_2, \beta_2, \gamma_2) = (2, 1, 1). \tag{3}$$

For arbitrary functions $h_i : \mathbb{Z}_{\geq 0} \to [-1, 1]$ and gains $a, b > 0$, set

$$f_1(p) = 1 + \epsilon a h_2(p), \qquad f_2(p) = 2 - \epsilon b h_1(p), \qquad g = \max(a, b). \tag{4}$$

The sign convention places no monotonicity constraint on $h_i$. For $\epsilon g < 1$, both rates are positive and bounded at every count.

These conditions define an admissible infinite-state process. Its total jump rate grows at most linearly, so it is nonexplosive. Positive immigration, translation and death allow communication between all states. For bounded transcription the Foster function

$$\Phi = 1 + \sum_{i=1}^{2} \left[ \left( 1 + \frac{1}{\delta_i} \right) M_i + P_i \right] \tag{5}$$

has negative linear drift outside a finite set. Polynomial powers of $\Phi$ have the corresponding negative leading drift, because jump increments are bounded and jump rates grow at most linearly. This gives positive recurrence, a unique stationary law $\pi_\epsilon$ and the polynomial moments needed below. Protein means are strictly positive.

Write $\mu_i = \mathbb{E}_\epsilon P_i$ and $F_i = \mathrm{Var}_\epsilon(P_i)/\mu_i$. The comparison gene has exactly the same $\delta_i, \beta_i, \gamma_i$ and constant transcription chosen to match $\mu_i$. Its Fano factor is Eq. (1), independent of that constant rate. Thus our baselines are

$$B_1 = \frac{3}{2}, \qquad B_2 = \frac{4}{3}. \tag{6}$$

They refer to the actual regulated means, not only to the parent means.

**Proposition 1** (Exact finite-lifetime noise identity)**.** *For the bounded reaction model* (2)*, with any finite positive kinetic coefficients,*

$$F_i - B_i = \frac{\beta_i \, \mathrm{Cov}_\epsilon(Z_i, f_i)}{\gamma_i \mu_i (\delta_i + \gamma_i)}, \qquad Z_i = \frac{\beta_i}{\delta_i} M_i + P_i. \tag{7}$$

*Proof.* Stationary generator tests on $M_i, P_i, M_i^2, M_iP_i, P_i^2$ give

$$\begin{aligned}
\delta_i \mathbb{E}M_i &= \mathbb{E}f_i, & \gamma_i \mathbb{E}P_i &= \beta_i \mathbb{E}M_i,\\
\mathrm{Var}(M_i) &= \mathbb{E}M_i + \frac{\mathrm{Cov}(M_i, f_i)}{\delta_i},\\
\mathrm{Cov}(M_i, P_i) &= \frac{\beta_i \,\mathrm{Var}(M_i) + \mathrm{Cov}(P_i, f_i)}{\delta_i + \gamma_i},\\
\mathrm{Var}(P_i) &= \mathbb{E}P_i + \frac{\beta_i}{\gamma_i}\,\mathrm{Cov}(M_i, P_i).
\end{aligned} \tag{8}$$

Substitution proves Eq. (7). The polynomial moment bounds justify the generator tests by localization. □

In the fixed units (3), $Z_1 = M_1 + P_1$, $Z_2 = M_2/2 + P_2$, and

$$F_i - B_i = \frac{\mathrm{Cov}_\epsilon(Z_i, f_i)}{(\delta_i + 1)\mu_i}. \tag{9}$$

In particular, the sign of the excess Fano factor is the sign of the corresponding covariance.

## 3 The local result

At $\epsilon = 0$, the two genes are independent constitutive parents, with $(c, \delta, \beta, \gamma) = (1, 1, 1, 1)$ and $(2, 2, 1, 1)$. Both RNA and protein means are one. Let $\mathcal{T}_i(t)$ be the single-parent backward semigroup and $\mathbb{E}_0$ expectation under the independent stationary parents. Define

$$\begin{aligned}
A_i &= \mathrm{Cov}_0(M_i, h_i(P_i)), & H_i &= \mathrm{Cov}_0(P_i, h_i(P_i)),\\
C_i(t) &= \mathrm{Cov}_0(h_i(P_i(0)), h_i(P_i(t))),\\
D_i(t) &= \mathbb{E}_0\big[(\mathcal{T}_i(t)h_i)(M_i + 1, P_i) - (\mathcal{T}_i(t)h_i)(M_i, P_i)\big].
\end{aligned} \tag{10}$$

The last expression is an extra-RNA response, not a simultaneous RNA/protein jump. Put

$$r_1(t) = (1 + t)e^{-t}, \qquad r_2(t) = e^{-t} - \tfrac{1}{2}e^{-2t}, \tag{11}$$

and define the exact injection and return forms

$$\begin{aligned}
U_1 &= \int_0^\infty r_1(t)C_2(t)\,\mathrm{d}t, & U_2 &= \int_0^\infty r_2(t)C_1(t)\,\mathrm{d}t,\\
V_1 &= A_1 \int_0^\infty r_1(t)D_2(t)\,\mathrm{d}t + H_1 \int_0^\infty e^{-t}D_2(t)\,\mathrm{d}t,\\
V_2 &= A_2 \int_0^\infty r_2(t)D_1(t)\,\mathrm{d}t + H_2 \int_0^\infty e^{-t}D_1(t)\,\mathrm{d}t.
\end{aligned} \tag{12}$$

Global additive constants in the responses do not affect these forms. The integrals are finite for bounded $h_i$.

**Theorem 1** (Local finite-RNA output-noise tradeoff)**.** *For the fixed parents above, every pair of bounded real whole-lattice responses satisfies*

$$|V_1|,\ |V_2| \ \le\ \frac{4}{5}\sqrt{U_1U_2}. \tag{13}$$

*For the physical family (4), define $S = a^2U_1 + b^2U_2$. If at least one $h_i$ is nonconstant, then $S > 0$. For every*

$$0 < \epsilon < \min\left\{\frac{1}{g}, \frac{S}{4480g^3}\right\}, \tag{14}$$

*the admissible stationary network has*

$$F_1 > \frac{3}{2} \quad \text{or} \quad F_2 > \frac{4}{3}. \tag{15}$$

*If both response functions are constant, $F_1 = 3/2$ and $F_2 = 4/3$.*

The operator inequality (13) is computer-assisted: its finite matrices, rational witnesses and infinite-state error bounds are described in Sec. 4 and Appendix B. The statement has no finite-plateau, smoothness or monotonicity restriction. It has a response-dependent amplitude: nearly constant or rare-tail responses can make $S$ arbitrarily small.

### 3.1 Finite-amplitude response and deduction

The independent-product Poisson equation for $(Z_i - \mathbb{E}_0 Z_i)(h_j - \mathbb{E}_0 h_j)$ gives the quadratic covariance terms in Eq. (12). A stationary coupling and weighted increment estimate supply the following finite remainder, proved in Appendix C:

$$\begin{aligned} \left|\mathrm{Cov}_\epsilon(Z_1, f_1) - \epsilon^2(a^2U_1 - abV_1)\right| &\le 448\epsilon^3 g^3, \\ \left|\mathrm{Cov}_\epsilon(Z_2, f_2) - \epsilon^2(b^2U_2 - abV_2)\right| &\le 448\epsilon^3 g^3. \end{aligned} \tag{16}$$

These are whole-lattice inequalities for $\epsilon g < 1$; stationary differentiability is not assumed.

Using Eq. (13) and $2ab\sqrt{U_1U_2} \le S$,

$$S - ab(V_1 + V_2) \ge \frac{1}{5}S. \tag{17}$$

Hence

$$\mathrm{Cov}_\epsilon(Z_1, f_1) + \mathrm{Cov}_\epsilon(Z_2, f_2) \ge \epsilon^2\frac{S}{5} - 896\epsilon^3 g^3 > 0 \tag{18}$$

under Eq. (14). At least one covariance is positive, and Eq. (9) proves Eq. (15). Strict positivity of $S$ follows from the weighted estimates below. When both responses are constant, the genes remain independent constitutive systems with changed means, so their Fano factors remain Eq. (6).

## 4 Controlling all response shapes

### 4.1 Injection variance and a weighted generator estimate

For the parent with RNA decay $d \in \{1, 2\}$, let $\pi_d$ denote its stationary law and $L_d$ its full backward generator. Its protein marginal has full support. For a bounded $h$, write $\widetilde{h} = h - \mathbb{E}_{\pi_d} h$ and

$$W_{4,d}(h) = \mathbb{E}_{\pi_d}\frac{\widetilde{h}(P)^2}{(1+P)^4}. \tag{19}$$

The scalar stationary covariance is even in time, although the parent process is not reversible. The impulse-filter covariance identity gives

$$U_1 = 2\,\mathrm{Var}_{\pi_2}\left(\int_0^\infty te^{-t}\widetilde{h}_2(P_t)\,\mathrm{d}t\right),$$
$$U_2 = 3\,\mathrm{Var}_{\pi_1}\left(\int_0^\infty (e^{-t}-e^{-2t})\widetilde{h}_1(P_t)\,\mathrm{d}t\right). \tag{20}$$

One can verify these identities by integrating the autocorrelations of the displayed kernels. Bounded responses justify Fubini. Conditioning on $(M_0, P_0)$ and applying Jensen yields

$$U_1 \geq 2\left\|(1-L_2)^{-2}\widetilde{h}_2\right\|_2^2,$$
$$U_2 \geq 3\left\|(1-L_1)^{-1}(2-L_1)^{-1}\widetilde{h}_1\right\|_2^2. \tag{21}$$

All norms are under the appropriate full invariant law.

Let $q_d(m,p) = d + (d+1)m + p$ be the total outgoing rate and $w_d = 1 + q_d$. Neighboring states have $w_d$ ratios between $1/2$ and $2$. Stationary incoming flux is $\pi_d(y)q_d(y)$. Outgoing-jump Cauchy–Schwarz, followed by this incoming-flux identity, gives

$$\left\|w_d^{-j}L_d u\right\|_2^2 \leq 2(1+2^{2j-1})\left\|w_d^{-(j-1)}u\right\|_2^2, \qquad j = 1, 2. \tag{22}$$

For the incoming term, the weight ratio costs at most $2^{2j-1}$; the outgoing diagonal term costs at most one. The factor two comes from $(u(y)-u(x))^2 \leq 2u(y)^2 + 2u(x)^2$. This uses nonnegative countable sums, not detailed balance. Applying Eq. (22) twice gives

$$\left\|w_d^{-2}(1-L_d)^2 v\right\|_2 \leq (1+2\sqrt{6}+\sqrt{108})\,\|v\|_2 < 17\,\|v\|_2\,,$$
$$\left\|w_d^{-2}(1-L_d)(2-L_d)v\right\|_2 \leq (2+3\sqrt{6}+\sqrt{108})\,\|v\|_2 < 20\,\|v\|_2\,. \tag{23}$$

The resolvent vectors in Eq. (21) have bounded first and second generator powers, so their backward resolvent equations justify these operations without a finite-support-core assumption.

The stationary cluster construction in Appendix A implies $\mathbb{E}[M \mid P = p] \leq p + 1$. Thus $\mathbb{E}[w_d \mid P = p] \leq 2(d+1)(p+1)$. Conditional Jensen for $x^{-4}$, Eq. (21) and Eq. (23) give

$$U_1 \geq \frac{W_{4,2}(h_2)}{187272}, \qquad U_2 \geq \frac{3W_{4,1}(h_1)}{102400}. \tag{24}$$

The denominators follow from $2/(17^2\,6^4) = 1/187272$ and $3/(20^2\,4^4) = 3/102400$. Every nonconstant bounded response therefore has strictly positive injection. The weight is deliberately weak; no stronger unproved coercivity estimate is needed.

### 4.2 Finite certificates and omitted functionals

Let $\mathcal{H}_d$ be the completion of bounded protein functions modulo constants under injection norm $U_2$ for $d = 1$, or $U_1$ for $d = 2$. Write the lower constants in Eq. (24) as $\alpha_1 = 3/102400$, $\alpha_2 = 1/187272$. For a zero-sum coefficient functional $\ell(h) = \sum_p \ell_p h(p)$,

$$\|\ell\|_{\mathcal{H}_d^*}^2 \leq \frac{1}{\alpha_d}\sum_{p\geq 0}\frac{(p+1)^4\ell_p^2}{\pi_d(P=p)}. \tag{25}$$

This follows by centering $h$ and applying Cauchy–Schwarz.

The return forms have rank two. With $D_{g,i} = \int e^{-t} D_i(t)\,\mathrm{d}t$ and $D_{r,i} = \int r_i(t) D_i(t)\,\mathrm{d}t$,

$$V_1 = A_1 D_{r,2} + H_1 D_{g,2}, \qquad V_2 = D_{r,1} A_2 + D_{g,1} H_2. \tag{26}$$

Appendix A provides whole-lattice coefficient bounds $|\ell_p| \le C(p+1)\pi_d(P=p)$ for each of these functionals. They are consequently continuous in $\mathcal{H}_d$.

Let $\mathcal{S}_n$ consist of response functions constant for $p \ge n$, modulo global constants, and let $\mathcal{P}_{d,n}$ be its injection-orthogonal projection. A full threshold category is included; this is not a cap on the stochastic process. Exact interval Gram matrices and integer congruence certificates at $n = 64$ establish

$$|V_i(\mathcal{P}_{1,n}h_1, \mathcal{P}_{2,n}h_2)| \le \frac{3}{4}\sqrt{U_2(h_1)U_1(h_2)}, \qquad n \le 64. \tag{27}$$

The parent resolvents are approximated on $0 \le M \le 48$, $0 \le P \le 128$ only to construct witnesses. Their errors under the true invariant law, including the infinite tail, enter every matrix enclosure. Appendix B specifies that error control.

For each return functional, a rational vector $v \in \mathcal{S}_n$ approximates its Riesz representative. Write its residual as $e = \ell - U_d(v, \cdot)$, where $U_d$ denotes the corresponding polarized injection form. In the scaled basis $s_k \mathbf{1}_{\{P=k\}}$ for $1 \le k < n$ and $s_n \mathbf{1}_{\{P \ge n\}}$, let $\rho_k$ enclose $|e(b_k)|$. The count-zero coefficient is bounded by $\sum_k \rho_k / s_k$, because $e$ annihilates constants and the last basis function covers the entire tail. The head dual bound is

$$H = \frac{(\sum_{k=1}^{n} \rho_k/s_k)^2}{\pi^-_{d,0}} + \sum_{p=1}^{n-1} \frac{(p+1)^4 (\rho_p/s_p)^2}{\pi^-_{d,p}}, \tag{28}$$

where $\pi^-_{d,p} > 0$ is a certified lower probability bound.

If $V \ge \sup_p |v(p) - \mathbb{E}v|$, the coefficient of the injection covariance operator obeys $|(U_d v)_p| \le c_d V \pi_{d,p}$, with $c_1 = 3/4$ and $c_2 = 2$. The omitted dual contribution is therefore at most

$$\begin{aligned} T &= C^2 T_6 + 2C c_d V T_5 + (c_d V)^2 T_4, \\ T_k &\ge \mathbb{E}_{\pi_d}[(P+1)^k \mathbf{1}_{\{P \ge n\}}], \qquad k = 4, 5, 6. \end{aligned} \tag{29}$$

Exact full moments minus lower probability prefixes enclose these positive tails. Thus

$$\Delta_\ell = \sqrt{(H+T)/\alpha_d} \tag{30}$$

is an upper bound on the full dual residual and on the norm of the omitted functional $\ell - \ell\mathcal{P}_{d,n}$. An upper bound for the projected functional is $m_\ell = \|v\|_{U_d} + \Delta_\ell$. For a factor pair $\ell, k$ in Eq. (26), its omitted operator norm is bounded by

$$\Delta_\ell m_k + m_\ell \Delta_k + \Delta_\ell \Delta_k. \tag{31}$$

Adding the two rank-one errors to 3/4 bounds the full return norm. No finite projection is assumed to exhaust the response space.

The decisive $n = 48$ exact bounds in Table 1 are strictly below 4/5. They prove Eq. (13) for all bounded response shapes. The smaller projection gives valid bounds below one but insufficient to certify 4/5; it is included to show the effect of the residual enclosure. Floating solves propose witnesses; exact rational residuals, probability intervals, moment tails and integer congruence margins decide all displayed inequalities.

| Riesz dimension $n$ | Return form | Full-space norm upper bound | Below 4/5 |
|---|---|---|---|
| 32 | $V_1$ | 0.806289935961 | No |
| 32 | $V_2$ | 0.846309121345 | No |
| 48 | $V_1$ | 0.750000465739 | Yes |
| 48 | $V_2$ | 0.750001582267 | Yes |

Table 1: Whole-space return bounds after finite Riesz projection and explicit residual-tail control. Each decimal is rounded upward from an exact rational bound. The finite projected norm bound is 3/4, certified in a 64-dimensional response space. The 32-dimensional projection is insufficient for the selected 4/5 constant; the 48-dimensional projection establishes it.

## 5 Physical interpretation and limitations

The positive injection forms measure fluctuations introduced by the other gene's transcription signal after RNA-to-protein filtering. The return terms measure how the loop acts back on that noise. At the stated low-copy parents, the full return response is too weak to offset both injection costs locally, even if the response is nonlinear, nonmonotone or concentrated on very rare protein counts. Finite RNA memory is retained throughout.

The result restricts a design route rather than asserting a universal noise law. It does not resolve simultaneous protein-noise suppression for arbitrary kinetic parameters, strong feedback or more than two genes. It also supplies no response-independent positive $\epsilon$ radius. The conservative constants are sufficient bounds, not optimal feedback thresholds or experimental predictions. Other channels, including promoter switching, resource coupling, sequestration and degradation control, lie outside Eq. (2). A comparison with a burst baseline at finite RNA lifetime would answer a different question.

The distinction from prior exact unit-birth and geometric-burst results is the explicit finite-RNA, output-only, mean-matched comparison. The literature comparison here is scoped to those primary models and the analytical regimes of Ref. [3]; it is not a claim to have excluded every representation of a related result.

## Data and computational methods

The companion source package contains the exact matrices, probability intervals, rational witnesses, residual bounds and certificate-replay code. The enclosure construction used Python with NumPy and SciPy for proposing sparse-resolvent and Riesz approximants; all final certificate signs use integer or `Fraction` arithmetic. A standard-library checker replays the stored contraction and residual certificates. Appendices below specify the underlying infinite-state arguments. There are no experimental data. The local package has no public repository identifier at this draft stage.

OpenAI Codex was used substantively in exploring the argument, literature synthesis, derivations, code generation, automated computational checks and manuscript drafting. The automated checks describe reproducible calculations, not completed human verification. Human scientific review and approval remain pending for this local draft. The accompanying author handoff records the disclosure details and model-version provenance that require confirmation before submission.

# A Exact parent laws and return coefficients

For $c = \delta = d \in \{1, 2\}$ and $\beta = \gamma = 1$, RNA immigrants form a Poisson process of rate $d$. Each RNA has an independent exponential lifetime of rate $d$, translates at rate one while alive, and gives proteins with independent exponential lifetimes of rate one. This labelled construction provides a positive stationary law on the infinite lattice.

The number of alive RNAs at observation is Poisson with mean one. An alive RNA has age distributed as $\mathrm{Exp}(d)$, so the mean number of its surviving proteins is $v = 1 - e^{-\mathrm{age}}$ with density $d(1-v)^{d-1}$ on $[0, 1]$. Its mark distribution is

$$q_d(k) = \frac{d}{k!} \int_0^1 e^{-v} v^k (1-v)^{d-1} \, \mathrm{d}v. \tag{32}$$

Dead RNAs supply independent compound-Poisson protein counts with Lévy coefficients and total intensity

$$\begin{aligned} \lambda_d(k) &= \frac{d}{k!} \int_0^1 e^{-v} v^{k-1} (1-v)^d \, \mathrm{d}v, \qquad k \geq 1, \\ \Lambda_d &= d \int_0^1 \frac{(1-e^{-v})(1-v)^d}{v} \, \mathrm{d}v. \end{aligned} \tag{33}$$

The joint stationary generating function is consequently

$$G_d(x, y) = \exp\left[ x q_d(y) - 1 + \sum_{k \geq 1} \lambda_d(k)(y^k - 1) \right], \tag{34}$$

where $q_d(y) = \sum_k q_d(k) y^k$. For $z = y - 1$,

$$\begin{aligned} q_d(1+z) &= d! \sum_{k \geq 0} \frac{z^k}{(d+k)!}, \\ \log G_{\mathrm{dead},d}(1+z) &= d \, d! \sum_{k \geq 1} \frac{z^k}{k(d+k)!}. \end{aligned} \tag{35}$$

These expressions obey $z q_d' = d + (z - d) q_d$ and $z(\log G_{\mathrm{dead},d})' = d(q_d - 1)$. Substitution in the full stationary forward equation

$$d(x-1)G + [d(1-x) + x(y-1)]G_x + (1-y)G_y = 0 \tag{36}$$

checks the defining reaction law, including catalytic translation. The known constitutive moments are

$$\mathbb{E}M = \mathbb{E}P = 1, \quad \mathrm{Var}(M) = 1, \quad \mathrm{Cov}(M, P) = \frac{1}{d+1}, \quad \mathrm{Var}(P) = 1 + \frac{1}{d+1}. \tag{37}$$

They reproduce Eq. (1); they are inputs, not the noise-tradeoff contribution.

Let $\pi_{d,p}$ be the protein marginal. It is compound Poisson with coefficients $\ell_k = q_d(k) + \lambda_d(k)$ for $k \geq 1$:

$$\pi_{d,0} = e^{-1 + q_d(0) - \Lambda_d}, \qquad \pi_{d,p} = \frac{1}{p} \sum_{k=1}^{p} k \ell_k \pi_{d,p-k}. \tag{38}$$

No higher probability coefficient enters this recurrence for a lower coefficient. All probabilities are positive. The alive-RNA Poisson Palm identity gives $\mathbb{E}[M \mathbf{1}_{\{P=p\}}] = (\pi_d * q_d)_p$. Alive RNAs with zero protein marks form an independent Poisson population of mean $q_d(0) \leq 1$. Each other alive RNA accounts for at least one protein, proving $\mathbb{E}[M \mid P = p] \leq p + 1$.

### A.1 Extra-RNA response

Add one fresh independently marked RNA at time zero to a stationary parent, and let $K_t$ be its surviving protein count at time $t$. Linearity of the parent channels gives

$$D_d(t;h) = \mathbb{E}[h(P+K_t) - h(P)], \tag{39}$$

where $P$ is stationary and independent of the added family. Conditionally on the extra RNA lifetime, $K_t$ is Poisson with mean between zero and one. For $k \geq 1$, direct mark integration gives

$$J_k(s) := \int_0^\infty e^{-st}\mathbb{E}[(K_t)_k]\,\mathrm{d}t = \frac{k!}{(s+k)(s+d)^{\overline{k}}}, \qquad J_0(s) = \frac{1}{s}. \tag{40}$$

Here $(x)_k$ is a falling factorial and $(s+d)^{\overline{k}} = \prod_{j=0}^{k-1}(s+d+j)$. Differentiation gives

$$-J_k'(s) = J_k(s)\left[\frac{1}{s+k} + \sum_{j=0}^{k-1}\frac{1}{s+d+j}\right]. \tag{41}$$

Define measures $\nu_s(k) = \int e^{-st}\Pr(K_t = k)\,\mathrm{d}t$ and $\nu_{s,t}(k) = \int te^{-st}\Pr(K_t = k)\,\mathrm{d}t$. Their coefficient expansion is

$$\nu_s(k) = \sum_{l\geq 0}\frac{(-1)^l\binom{k+l}{k}}{(s+k+l)(s+d)^{\overline{k+l}}}, \tag{42}$$

with each term for $\nu_{s,t}(k)$ additionally multiplied by the bracket in $-J_{k+l}'(s)/J_{k+l}(s)$. Even and odd Taylor sums for the conditional Poisson factor $e^{-v_t}$ enclose these integrals because $0 \leq v_t \leq 1$.

The coefficient sequences of the return functionals are

$$\begin{aligned} A_p &= (\pi_d * q_d)_p - \pi_{d,p}, & H_p &= (p-1)\pi_{d,p}, \\ (D_s)_p &= (\pi_d * \nu_s)_p - \pi_{d,p}/s, & (D_{s,t})_p &= (\pi_d * \nu_{s,t})_p - \pi_{d,p}/s^2. \end{aligned} \tag{43}$$

All have zero total sum. In Eq. (26), $D_{g,d} = D_1$; $D_{r,1} = D_1 - \frac{1}{2}D_2$ and $D_{r,2} = D_1 + D_{1,t}$.

For $k \geq 1$, $q_d(k) \geq e^{-1}d!/(d+k)!$. Comparing this with $J_k(s)/k!$, and using $e < 3$, shows $\nu_1(k), \nu_2(k), \nu_{1,t}(k) \leq (3/2)q_d(k)$. For the time-weighted case the extra harmonic bracket is at most $(k+1)/2$. Splitting off count zero in the convolutions, using total measure mass at most one and the conditional RNA bound, yields

$$|\ell_p| \leq C_\ell(p+1)\pi_{d,p}, \qquad \begin{array}{c|cccc} \ell & A & H & D_g & D_r \\ \hline d=1 & 1 & 1 & 5/2 & 15/4 \\ d=2 & 1 & 1 & 5/2 & 5 \end{array} \tag{44}$$

The tabulated entries are $C_\ell$. Together with Eq. (25), these inequalities control each functional on the entire response space.

## B Infinite-state numerical enclosures

### B.1 Probabilities, moments and resolvent residuals

The stationary cluster integrals are enclosed by the 160th and 161st alternating Taylor sums, with outward fixed-point arithmetic at 384 bits. Positive convolution recurrences evaluate the joint law (34) and marginal (38). The joint calculation uses

$$\pi_d(m,p) = \frac{e^{-1}}{m!}[y^p]q_d(y)^m G_{\mathrm{dead},d}(y). \tag{45}$$

No finite stationary solve or artificial normalization supplies its probabilities.

For $f_{r,s} = \mathbb{E}[(M)_r (P)_s]$, direct generator testing gives

$$(rd+s) f_{r,s} = rd\, f_{r-1,s} + s\big[f_{r+1,s-1} + r f_{r,s-1}\big], \qquad f_{0,0} = 1. \tag{46}$$

The recurrence is triangular at each total degree, proceeding from smaller $s$. Exact mixed factorial moments through degree 104 bound the required tails. For example, factorial Markov bounds at $R+1$ and $S+1$, together with

$$\begin{aligned} M(M)_k &= (M)_{k+1} + k(M)_k, \\ M^2(M)_k &= (M)_{k+2} + (2k+1)(M)_{k+1} + k^2 (M)_k, \end{aligned} \tag{47}$$

provide upper bounds $\tau_j$ on $\mathbb{E}[M^j \mathbf{1}_{\{M>R \text{ or } P>S\}}]$ for $j = 0, 1, 2$. The $P$-tail terms use the corresponding mixed moments. Positive bounds are minimized over available factorial orders $k \le 100$ without moment closure.

Construct approximate resolvent vectors on $R = 48, S = 128$ using the capped backward generator. Quantize them as integers over $2^{192}$, then extend them to the full lattice by coordinate clipping. This cap defines a witness construction, not the stationary process. For each source category, the source itself equals its clipped extension outside the rectangle. If $\eta$ is the maximum exact interior residual and $V$ the approximant sup norm, then the extended residual obeys

$$|r(m,p)| \le \eta + 2V[dR + S + (d+1)m] \mathbf{1}_{\{m>R \text{ or } p>S\}}. \tag{48}$$

Missing capped death terms and excess translation account for this bound. Clipping in $p$ prevents an unbounded $p$ coefficient. With $A = dR + S$ and $B = d+1$,

$$\begin{aligned} A_{\text{tail}} &= A^2 \tau_0 + 2AB\tau_1 + B^2 \tau_2, \\ \|r\|_2^2 &\le \sum_{\text{inside}} \pi_d^+(m,p) r(m,p)^2 + 2\eta^2 \tau_0 + 8V^2 A_{\text{tail}}. \end{aligned} \tag{49}$$

Only exact nonnegative sums and positive tail bounds enter this enclosure.

Conditional Jensen and invariance make the full Markov semigroup an $L^2(\pi_d)$ contraction. Hence $\|(s - L_d)^{-1}\|_{2\to 2} \le 1/s$. For the exact resolvent $v_s$ and its clipped approximant $\widehat{v}_s$,

$$\|v_s - \widehat{v}_s\|_2 \le \|r_s\|_2 / s. \tag{50}$$

For $w = (1 - L_d)^{-2} b$, whose approximant uses $\widehat{v}_1$ as source,

$$\|w - \widehat{w}\|_2 \le \|r_1\|_2 + \|r_w\|_2 . \tag{51}$$

Localization and finite polynomial moments justify these residual identities for the clipped vectors.

Use disjoint protein categories $1, \dots, n-1$ and $\{P \ge n\}$, omitting zero modulo constants. Scale each by an exact power of two $s_k$ so that $1/4 < s_k^2 \pi_k^+ \le 1$. Testing a residual against a centered scaled category costs at most one in $L^2$. The injection-matrix resolvent errors are therefore controlled by $2\|r_1\|_2 + \|r_w\|_2$ for $U_1$, and $\|r_1\|_2 + \|r_2\|_2 /4$ for $U_2$. The second expression includes the $-1/2$ kernel coefficient and the $1/2$ resolvent norm. Probability integration and centering errors are propagated separately before symmetrization.

Return matrices use the one-dimensional coefficient sequences (43). The full threshold coefficient is the negative prefix sum through $p = n-1$, since each sequence has zero total sum. No derivative at an RNA boundary is used.

### B.2 Exact block positivity

Let $G_2, G_1$ be the enclosed injection matrices on $\mathcal{S}_{64}$ and $\mathsf{B}$ the matrix for $V_1$ or $V_2$ in the corresponding $(h_1, h_2)$ ordering. The required block is

$$K = \begin{pmatrix} (3/4)G_2 & \mathsf{B} \\ \mathsf{B}^{\mathsf{T}} & (3/4)G_1 \end{pmatrix}. \tag{52}$$

A floating Cholesky inverse proposes a congruence $T$, then its entries are quantized as integers over $2^{96}$. The central block $K_0$ is stored as integers over $2^{192}$; its rounding is added to a common true-entry error bound $e$. If $\ell_i$ is the absolute row sum of the rational $T$, each transformed entry error is at most $e\ell_i\ell_j$. The exact sufficient margins are

$$(TK_0T^{\mathsf{T}})_{ii} - \sum_{j\neq i} |(TK_0T^{\mathsf{T}})_{ij}| - e\ell_i \sum_j \ell_j > 0. \tag{53}$$

The two 128-dimensional blocks have all these rational margins positive; their minimum margins both exceed 0.9948. Symmetric strict diagonal dominance gives positivity of the true transformed block. It also forces $T$ to have full rank, proving positivity of the true original block. Rescaling the two input vectors in Eq. (52) then gives Eq. (27). No eigenvalue rounded to a positive decimal decides this test.

### B.3 Riesz residuals and full return norms

Finite Riesz vectors at $n = 32, 48$ are proposed by a floating solve, then quantized to integers over $2^{128}$. The $n = 64$ Gram enclosures are transformed by an exact basis map to the smaller spaces; the new threshold is the sum of all old threshold categories. Source intervals and Gram errors enclose every scaled residual in Eq. (28).

For completeness, the polarized injection operator has coefficient

$$(U_d v)_p = \frac{\pi_{d,p}}{2} \int_0^\infty r_{\mathrm{rec}}(t)\, [\mathbb{E}[v(P_t) - \mathbb{E}v \mid P_0 = p] + \mathbb{E}[v(P_0) - \mathbb{E}v \mid P_t = p]]\, \mathrm{d}t, \tag{54}$$

where $r_{\mathrm{rec}} = r_2$ for $d = 1$, and $r_1$ for $d = 2$. Both conditional expectations have absolute value at most $V$. The kernel masses are $3/4$ and 2, giving the bound used in Eq. (29) without assuming reversibility.

Full moments for the tail subtraction can also be reconstructed from the exact protein factorial cumulants

$$\kappa_k = \frac{(k-1)!\, d!}{(d+k-1)!}. \tag{55}$$

The Bell recurrence gives factorial moments, and Stirling conversion gives $\mathbb{E}[(P+1)^j]$. Subtracting the lower probability prefix yields the upper $T_j$ for $j = 4, 5, 6$. This is an independent moment representation of the same parent law.

All residual products and tail subtractions are rational. Square-root upper bounds are obtained with integer square roots and checked by squaring. Orthogonal projection contracts the dual residual norm: Eq. (30) bounds $\ell - \ell\mathcal{P}_{d,n}$ even though the finite spaces need not exhaust $\mathcal{H}_d$. Equations (26) and (31) then give the full return bounds in Table 1. The stored rational data and standard-library replay code specify the complete finite arithmetic.

# C Finite-feedback covariance remainder

This appendix proves Eq. (16) without assuming a differentiable stationary distribution. All response functions here retain their original sup bound one, even when constants are removed for the response-space analysis.

## C.1 A stationary immigration-envelope coupling

For gene $i$, use common RNA immigration rate $c_i - \epsilon g$ and an additional envelope immigration process of rate $2\epsilon g$, where $(c_1, c_2) = (1, 2)$. Each arrival is independently marked with RNA lifetime, translation events and protein lifetimes. A constitutive parent $Y_i$ accepts each envelope arrival with probability $1/2$. The regulated process $X_i$ accepts it with probability

$$\frac{f_i - (c_i - \epsilon g)}{2\epsilon g} \in [0, 1], \tag{56}$$

using the same uniform acceptance mark. Acceptance depends on the current regulated state, but the descendant marks of the new family are independent of that decision. Let $E_i$ be the process retaining every envelope family. Then, pathwise,

$$|X_{M,i} - Y_{M,i}| \le M_{E,i}, \qquad |X_{P,i} - Y_{P,i}| \le P_{E,i}. \tag{57}$$

The parents remain independent across genes. The two envelopes are also independent, and $Y_i$ is independent of $E_j$ for $j \neq i$. The parent and its own envelope share the parent's half-envelope; their covariance must be retained. Stationary joint limits follow from constructions started in the past, uniform dominating polynomial moments and uniqueness of the marginal invariant laws. No monotone or associated feedback process is required.

For $W = M + P$ in a constitutive gene with transcription $c$ and $\delta = d \ge 1$, the exact moments give

$$\mathbb{E}W = \frac{2c}{d}, \qquad \mathrm{Var}(W) = \frac{c}{d}\left(2 + \frac{3}{d+1}\right). \tag{58}$$

Thus $\mathbb{E}W_{Y,i} = 2$, $\mathbb{E}W_{E,i} \le 4\epsilon g$ and $\mathrm{Cov}(W_{Y,i}, W_{E,i}) \le 7\epsilon g/2$. For either $i$,

$$\mathbb{E}[(1 + W_{Y,i})(W_{E,1} + W_{E,2})] \le \frac{55}{2}\epsilon g. \tag{59}$$

The constant is $3(4 + 4) + 7/2$; it includes the parent's dependence on its own envelope.

## C.2 Product-parent Poisson solution

Write $\eta_j = \mathbb{E}_0 h_j(P_j)$ and $\bar{z}_i = 1 + 1/\delta_i$. For $i \neq j$, let

$$\begin{aligned} q_i &= (Z_i - \bar{z}_i)(h_j(P_j) - \eta_j), \\ u_i(t, m_i, p_i) &= r_i(t)(m_i - 1) + e^{-t}(p_i - 1), \\ v_j(t, m_j, p_j) &= \mathcal{T}_j(t) h_j(m_j, p_j) - \eta_j, \\ Rq_i &= \int_0^\infty u_i(t, m_i, p_i) v_j(t, m_j, p_j)\, \mathrm{d}t. \end{aligned} \tag{60}$$

The affine parent mean evolution and independence give $L_0 Rq_i = -q_i$. Here $|v_j| \le 2$ and $r_i(t) \le (1 + t)e^{-t}$, so $Rq_i$ grows at most linearly and its generator increments have integrable polynomial

growth. Localization and the bounded-network moment estimates justify stationary generator testing. With

$$L_\epsilon = L_0 + \epsilon L_1, \qquad L_1 = a h_2(P_2)\Delta_{M_1} - b h_1(P_1)\Delta_{M_2}, \tag{61}$$

put $k_i = L_1 R q_i$. Then

$$\mathbb{E}_\epsilon q_i = \epsilon \mathbb{E}_\epsilon k_i, \qquad \mathbb{E}_0 k_1 = aU_1 - bV_1, \quad \mathbb{E}_0 k_2 = aV_2 - bU_2. \tag{62}$$

The two RNA birth channels acting separately on the product Poisson solution give the last two identities. This also shows why the Fano deviations start at quadratic order after multiplying by the signed transcription gain.

### C.3 Weighted increment estimate

Any function from integer counts to $[-1, 1]$ has discrete Lipschitz constant at most two. Coupling unregulated descendant families, for $d \geq 1$, gives

$$\begin{aligned} |\mathcal{T}_j(t)h_j(z) - \mathcal{T}_j(t)h_j(y)| &\leq 2e^{-t}(\Delta p_j + t\Delta m_j), \\ |\Delta_M \mathcal{T}_j(t)h_j(z)| &\leq 2te^{-t}, \\ |\Delta_M \mathcal{T}_j(t)h_j(z) - \Delta_M \mathcal{T}_j(t)h_j(y)| &\leq 4e^{-t}(\Delta p_j + t\Delta m_j), \end{aligned} \tag{63}$$

where $\Delta m_j = |m_j - m_j'|$, $\Delta p_j = |p_j - p_j'|$ and $y$ is the primed state. For the feedforward term in $k_i$, integrating the product difference costs at most $g[\Delta m_j + (19/2)\Delta p_j]$.

For the return product $h_i u_i \, \Delta_M \mathcal{T}_j h_j$, use the exact telescoping into the difference of $u_i$, of $h_i$, and of the extra-RNA response, with the other factors fixed in turn. The bound $|u_i(t, y)| \leq r_1(t)(W_i' + 2)$ and the integrals

$$\int r_1 = 2, \qquad \int r_1 e^{-t} = \frac{3}{4}, \qquad \int t r_1 e^{-t} = \frac{1}{2}, \qquad \int t e^{-2t} = \frac{1}{4} \tag{64}$$

give

$$g\left[\Delta m_i + \left(\frac{9}{2} + 2W_i'\right)\Delta p_i + (4 + 2W_i')\Delta m_j + (6 + 3W_i')\Delta p_j\right]. \tag{65}$$

Adding the feedforward bound therefore yields

$$\begin{aligned} |k_i(z) - k_i(y)| &\leq g\left[\Delta m_i + \left(\frac{9}{2} + 2W_i'\right)\Delta p_i + (5 + 2W_i')\Delta m_j + \left(\frac{31}{2} + 3W_i'\right)\Delta p_j\right] \\ &\leq 16g(1 + W_i')(\Delta m_i + \Delta p_i + \Delta m_j + \Delta p_j). \end{aligned} \tag{66}$$

Applying this to the stationary coupling and Eq. (59) gives

$$|\mathbb{E}_\epsilon k_i - \mathbb{E}_0 k_i| \leq 440\epsilon g^2. \tag{67}$$

Exact first-moment balance gives $|\mathbb{E}_\epsilon Z_i - \bar{z}_i| \leq 2\epsilon g$, and Eq. (57) gives $|\mathbb{E}_\epsilon h_j - \eta_j| \leq 4\epsilon g$. Subtracting the mean product in $\mathbb{E}_\epsilon q_i$, and using Eq. (62), proves

$$|\mathrm{Cov}_\epsilon(Z_i, h_j) - \epsilon \mathbb{E}_0 k_i| \leq 448\epsilon^2 g^2. \tag{68}$$

Multiplication by $\epsilon a$ for gene one and by $-\epsilon b$ for gene two proves Eq. (16). The estimate is specific to the fixed unequal pair and is not an arbitrary-kinetics remainder theorem.

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